Mayar (May) **Ahmed**

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Skills

Programming	Python, Bash, R, MATLAB; object-oriented design
Software Engineering	Git/GitHub, GitHub Actions, pytest, Ruff, pixi; Linux, SLURM, HPC
Cheminformatics	Docking (Glide, GNINA, AutoDock Vina), Foldseek, AlphaFold2; RDKit, PDB2PQR, Gypsum-DL, Open Babel
Genomics	Assembly (Flye), alignment (minimap2, BWA, samtools), variant calling (bcftools, bedtools); BLAST, BUSCO, CRAQ
Scientific Computing	NumPy, SciPy, pandas, Polars, PyArrow, Biopython, MDAnalysis
Machine Learning	PyTorch, scikit-learn; supervised learning, feature engineering, model evaluation, Random Forest

Research Experience

Graduate Student Researcher — Dr. Jacob D. Durrant

Department of Biological Sciences, University of Pittsburgh — *Sep 2021 – Present*

- Built **LIGNOVA**, an open-source Python/SLURM pipeline that docks PubChem bioactives into experimental PDB structures with GNINA on HPC, generating **258M+ docked poses** (243,805 ligands; 5.86M pairs, 89% affinity-labeled), each graded on pose and affinity confidence into Gold/Silver/Bronze tiers; dataset analysis exposed systematic biases in ML scoring functions.
- Developed **reqadence**, an open-source async Python REST-client base class powering LIGNOVA's RCSB/PubChem data ingestion.
- Benchmarked **consensus rescoring** (ComBind) vs. explicit water in **Glide** pose prediction. **Manuscript in preparation.**
- Led a structure-based reverse screen (**Foldseek & MolModa docking**) that prioritized 4 candidate arginine-sensing proteins in *S. cerevisiae* missed by sequence homology; experimental follow-up by the O'Donnell Lab confirmed phenotypes in a subset.

Undergraduate Student Researcher — Dr. Eman Badr

University of Science and Technology, Zewail City — *Jun 2020 – Apr 2021*

- Trained Random Forest models (scikit-learn) to predict drug sensitivity from transcriptomic profiles in AML patients; identified genetic markers correlated with differential drug response.

Education

Department of Biological Sciences, University of Pittsburgh

Ph.D. candidate in Molecular, Cellular, and Developmental Biology, Dr. Jacob Durrant — *Aug 2021 – Expected Apr 2027*

University of Science and Technology, Zewail City

B.S. in Computational Biology and Genomics (Biomedical Science) — *Aug 2017 – Jul 2021*

Publications

Holland, M.*, **Ahmed, M.***; Young, J. M.; Drurey, J. R.; McFadyen, S.; Ostrowski, E. A.; Levin, T. C. Hypermutable hotspot enables the rapid evolution of self/non-self recognition genes in *Dictyostelium*. *PNAS* **2025**, *122*(51). DOI: 10.1073/pnas.2520843122. (*co-first authors)

Kochnev, Y.; **Ahmed, M.**; Maldonado, A. M.; Durrant, J. D. MolModa: Accessible and secure molecular docking in a web browser. *Nucleic Acids Research* **2024**, *52*(W1), W498–W506. DOI: 10.1093/nar/gkae406.

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From byte to bench to bedside: Molecular dynamics simulations and drug discovery. *BMC Biology* **2023**, *21*(1), 299. DOI: 10.1186/s12915-023-01791-z.

Conferences

2026	**Poster Presenter**: LIGNOVA: An Automated Pipeline for Large-Scale Generation of High-Quality Protein–Ligand Complexes, Computational Medicinal Chemistry School	*Cambridge, MA*
2024	**Poster Presenter**: Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery, Gordon Research Conference & Seminar on Computational Chemistry	*Portland, ME*
2023	**Poster Presenter**: Augmenting Protein–Ligand Complex Databases with PubChem for Enhanced Drug Discovery, Gordon Research Conference & Seminar on Computer-Aided Drug Design	*West Dover, VT*

Selected Honors & Awards

2026	**NSF ACCESS Allocation (BIO260240)**, Competitive national computing allocation to scale LIGNOVA	*USA*
2021	**Provost's Honors Roll**, Zewail City — top 5 students by cumulative GPA	*Giza, Egypt*
2021	**Cum Laude**, B.S. Computational Biology & Genomics, Zewail City	*Giza, Egypt*